

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) — Semester IV Examination, 2013-14

Paper No. MTB-401: Partial Differential Equations

(Common for Old and New Course)

Full Marks: 70

Time: 3 Hours

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1.

- (a) Find the PDE by eliminating the arbitrary constants a and b from $ax^2 + by^2 + z^2 = 1$.
[2]
- (b) Eliminate the arbitrary function f from $z = f(x - y)$. [2]
- (c) Find the general integral of $xp + yq = z$. [2]
- (d) Find the complete integral of $pq = xy$. [2]
- (e) Find the complete integral of $pq = p + q$. [2]
- (f) Solve $2r + 5s + 2t = 0$. [2]
- (g) Solve $(D + D' - 1)(D + 2D' - 3)z = 0$. [2]
- (h) Solve $z = px + qy + \ln(pq)$. [2]
- (i) Solve $\frac{\partial^2 z}{\partial y^2} = \sin(xy)$ (i.e. $t = \sin xy$). [2]
- (j) Solve $xr + 2p = 0$. [2]
- (k) Solve $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial y}$ given $u(0, y) = 8e^{-3y}$. [2]

Question 2.

- (a) Find the general integral of $x^2p + y^2q = (x + y)z$. [6]
- (b) Find the complete integral of $yzp^2 - q = 0$. [6]

Question 3.

- (a) Solve: $4r - 4s + t = 16 \ln(x + 2y)$. [6]
- (b) Solve: $(D^2 + 2DD' + D'^2)z = 2 \cos y - x \sin y$. [6]

Question 4.

(a) Solve: $s + p - q = z + xy$ (i.e. $\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = z + xy$). [6]

(b) Solve: $(D^2 + DD' + D' - 1)z = \sin(x + 2y)$. [6]

Question 5.

(a) Solve: $ys + p = \cos(x + y) - y \sin(x + y)$. [6]

(b) Solve: $xr + ys + p = 10xy^2$. [6]

Question 6. Solve by Monge's method: (each 6 marks)

(a) $(q + 1)s = (p + 1)t$ [6]

(b) $3s + (rt - s^2) = 2$ [6]

Question 7.

(a) Obtain the general solution of the wave equation:

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$

[6]

(b) Solve the heat equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{1}{k} \frac{\partial u}{\partial t}$$

[6]

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